

7. There are two types of nuclear reaction: fission and fusion.

(a) Two isotopes of hydrogen ${}^2_1\text{H}$ and ${}^3_1\text{H}$ can fuse together to form helium (He).

(i) **Complete** the equation for this fusion reaction. [2]



(ii) **Complete** the table below to give the number of protons and neutrons in the nuclei of each hydrogen isotope. [4]

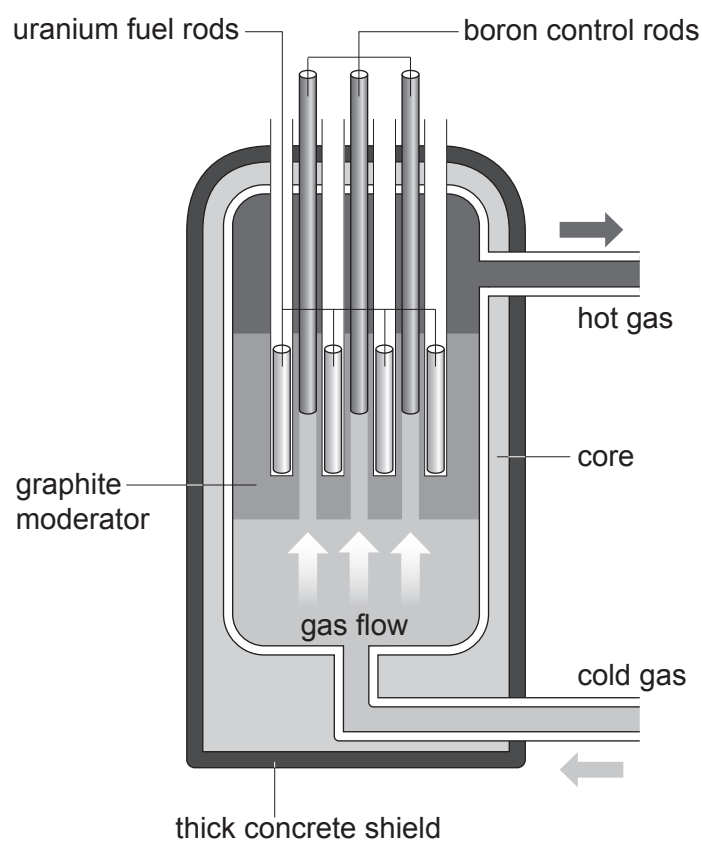
Isotope	${}^2_1\text{H}$	${}^3_1\text{H}$
number of protons		
number of neutrons		

(iii) State **two** reasons why nuclear fusion is difficult to achieve on Earth. [2]

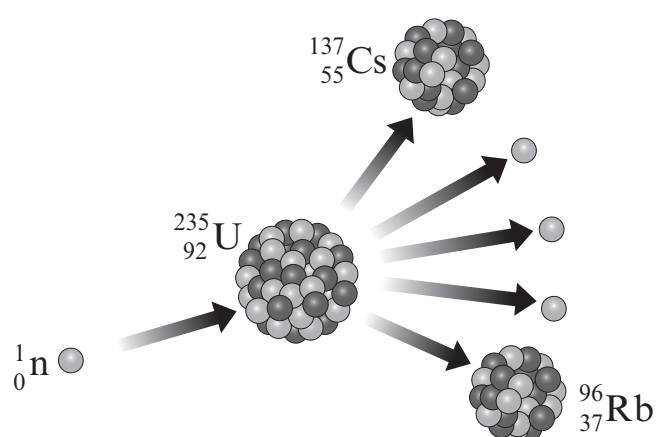
1.
2.



(b) Nuclear fission is used to produce electricity in power stations.



In a nuclear power station, a neutron is absorbed by a uranium nucleus, producing two lighter nuclei and more neutrons.
Part of the chain reaction is shown below.



- (i) State the number of neutrons produced in the fission reaction shown in the diagram. [1]

number of neutrons =

- (ii) State which part of a nuclear reactor slows down neutrons. [1]

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- (iii) State why neutrons need to be slowed down. [1]

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- (iv) State the purpose of the control rods. [1]

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